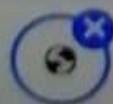


Select the correct option

- ☐ a) by subtracting the cumulative frequency of the relative class from the next cumulative frequency
- ☐ b) by subtracting the cumulative frequency of the relative class from the total frequency
- ☐ c) by subtracting the cumulative frequency of the relative class from the previous cumulative frequency
- ☐ None of these



If some values are given then their mean may be

.....

Select the correct option

☐ Negative

☐ Positive

☐ zero

☐ All above





BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 9 of 10 (Start time: 12:19:44 AM, 07 February 2021)

Total Marks: 1

Relative frequency is defined as

Select the correct option

- ☐ the ratio of frequency of the class and total frequency
- ☐ frequency obtained by adding the previous to the next class frequency
- ☐ the ratio of frequency of the class to the highest frequency
- ☐ None of these

[Click to Save Answer & Move to Next Question](#)



BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

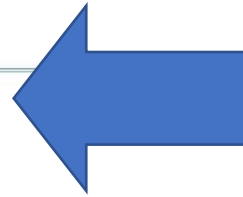
Question # 10 of 10 (Start time: 12:21:07 AM, 07 February 2021)

Total Marks: 1

For Symmetrical Distribution, if mean = 20, then -----

Select the correct option

- | | |
|-----------------------|------------------------|
| ☐ | mode = 19, median = 18 |
| ☐ | mode = 18, median = 18 |
| ☐ | mode = 20, median = 20 |
| ☐ | mode = 19, median = 19 |



Click to Save Answer & Move to Next Question



Question # 10 of 10 (Start time: 11:52:51 PM, 06 February 2021)

Which of the following is true?

Select the correct option

☐

$$A.M > G.M > H.M$$

☐

$$A.M < G.M < H.M$$

☐

$$A.M = G.M = H.M$$

☐

None of these.



If the value '2' collected in the data $n/2$ times and '3' occurs $n/2$ -time then the arithmetic mean is

Select the correct option

- ☐ $5n/2$
- ☐ $5/2$
- ☐ $5(n-1)/2$
- ☐ None of these





BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 1 of 10 (Start time: 12:11:52 AM, 07 February 2021)

Total Marks: 1

What is the median of this set of numbers: 4, 6, 7, 9, 2000000?

Select the correct option

☐	7.5
☐	6
☐	7
☐	4

Click to Save Answer & Move to Next Question



Question # 8 of 10 (Start time: 11:50:45 PM, 06 February 2021)

Linear Graphs are obtained from

Select the correct option

- ☐ linear equation
- ☐ Quadratic equations
- ☐ Cubic equation
- ☐ None of these





BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 2 of 10 (Start time: 12:12:25 AM, 07 February 2021)

Total Marks: 1

Another name of mean in statistics is

Select the correct option

- | | |
|-----------------------|----------------|
| ☐ | Mode |
| ☐ | Expected value |
| ☐ | Median |
| ☐ | None of these |



Click to Save Answer & Move to Next Question

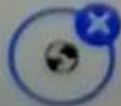


Question # 7 of 10 (Start time: 11:50:05 PM, 06 February 2021)

Total Marks: 1

The range of the data 2, 5, 8, 9, 15, 6, 7 is

Select the correct option

☐ 13☐ 11☐ 9☐ 10



BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

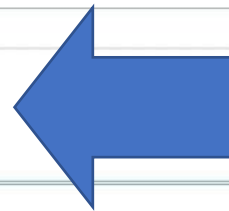
Question # 3 of 10 (Start time: 12:13:20 AM, 07 February 2021)

Total Marks: 1

Measures of central tendency are used to measure.....

Select the correct option

- ☐ location of middle value
- ☐ location of starting value
- ☐ location of end value
- ☐ none of these



Click to Save Answer & Move to Next Question



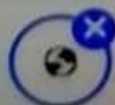
Question # 6 of 10 (Start time: 11:49:29 PM, 06 February 2021)

Total Marks: 1

Range of data: 4,3,2,1,0,-1,-2,-3,-4 is

Select the correct option

- ☐ -8
- ☐ 4
- ☐ -4
- ☐ 8





BC190200960: ZAID NASEEM MUGHAL

Time Left 90 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 4 of 10 (Start time: 12:14:15 AM, 07 February 2021)

Total Marks: 1

Linear Graphs are obtained from

Select the correct option

- ☐ linear equation
- ☐ Quadratic equations
- ☐ Cubic equation
- ☐ None of these

Click to Save Answer & Move to Next Question





What is the median of this set of numbers: 4, 6, 7, 9, 2000000?

Select the correct option

7.5



6



7



4





BC190200960: ZAID NASEEM MUGHAL

Time Left 89 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

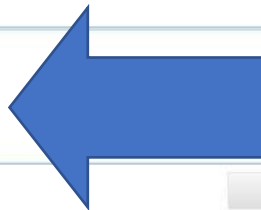
Question # 5 of 10 (Start time: 12:14:57 AM, 07 February 2021)

Total Marks: 1

If the minimum and maximum values for the given data is same than

Select the correct option

- ☐ mean would be zero
- ☐ mean would be maximum value
- ☐ mean less than minimum value
- ☐ none of these



Click to Save Answer & Move to Next Question





BC200381201: HAMID UL HASSAN

Time Left: 70 sec(s)

MTN302:quiz 2

Quiz Start Time: 11:45 PM, 06 February 2021

Question # 4 of 10 (Start time: 11:47:14 PM, 06 February 2021)

Total Marks: 1

$$H = \frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

is used to calculate _____

Select the correct option

[Related Math Equations](#)

- | | |
|-----------------------|-----------------|
| ☐ | Harmonic mean |
| ☐ | Geometric Mean |
| ☐ | Arithmetic Mean |
| ☐ | None of Above |

[Click on Next Answer & Uncheck Your Answer](#)



BC190200960: ZAID NASEEM MUGHAL

Time Left 89 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 6 of 10 (Start time: 12:16:17 AM, 07 February 2021)

Total Marks: 1

Frequency Distributions involve

Select the correct option

- | | |
|-----------------------|--------------|
| ☐ | Tables |
| ☐ | Histograms |
| ☐ | Polygons |
| ☐ | All of these |



Click to Save Answer & Move to Next Question



The statement

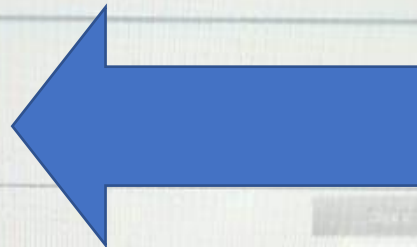
"Mean does not get affected by the extreme values or outliers in the data"
is

Select the correct option

True



False





BC190200960: ZAID NASEEM MUGHAL

Time Left 89 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 7 of 10 (Start time: 12:17:30 AM, 07 February 2021)

Total Marks: 1

If the mean of the data is 50 and sum of the data is 450 then the number of observations in the data will be

Select the correct option

☐	9
☐	500
☐	22500
☐	None of these

Click to Save Answer & Move to Next Question





BC190200960: ZAID NASEEM MUGHAL

Time Left 88 sec(s)

MTH302:quiz 2

Quiz Start Time: 12:11 AM, 07 February 2021

Question # 8 of 10 (Start time: 12:17:49 AM, 07 February 2021)

Total Marks: 1

The formula for Class Midpoint =

Select the correct option

- | | |
|-----------------------|---|
| ☐ | $(\text{Lower limit} + \text{Upper limit})/2$ |
| ☐ | $(\text{Lower limit} - \text{Upper limit})/2$ |
| ☐ | Both above |
| ☐ | none of these |

Click to Save Answer or Move to Next Question



MTH 302 QUIZ FILE # 6 Feb 2021

Updated on 2/6/2021 7:15 PM

Q.1 If the mean of the data is 50 and sum of the data is 450 then the number of observations in the data will be

A. 9 B. 500 C. 22500 D.
None of these

Q.2 The frequency of the required class can be obtained from the cumulative frequency

- A. by subtracting the cumulative frequency of the relative class from the next cumulative frequency
- B. by subtracting the cumulative frequency of the relative class from the total frequency
- C. by subtracting the cumulative frequency of the relative class from the previous cumulative frequency**
- D. None of the given

Q.3 Mode of the data 25,35,21,56,25,56,65 is.....

A. 35

B. 25 & 56

C. 21

D. 65

Q.4 In Positively Skewed Distribution, which of the following is true?

A. Mean > median > mode

B.

Mode > median > mean

C. Mean > mode > median

D.

None of the given

Positively Skewed Distribution
(Tilted to left)

mean > median > mode

Q.5 Last year a small statistical consulting company paid each of its five statistical clerks \$22,000, two statistical analysts \$50,000 each, and the senior statistician/owner \$270,000. The number of employees earning less than the mean salary is

A. 4

B. 5

C. 6

D. 7

Q.6 The most frequent/common value in the data is called

- A. Mean B. Median **C. Mode**
D. Midrange

Q.7 _____ is the mean of inverse values.

A. Harmonic mean

B.

Geometric mean

C. Arithmetic mean

D. Midhing

HARMONIC MEAN

Harmonic mean is defines as under:

$$HM = \frac{n}{(1/x_1 + 1/x_2 + \dots + 1/x_n)} = \frac{n}{\text{Sum}(1/x_i)}$$

Example

Find HM of 10, 8, 6

$$HM = \frac{3}{(1/10 + 1/8 + 1/6)} = 7.66$$

Q.8 Ogives are used to determine _____
at different values.

- A. Frequency
- B. Frequency distribution
- C. The cumulative Frequency**
- D. None of these

Q.9 The measure of central tendency which divides the data into two equal parts is called

- A. Mode **B. Median** C.
Mean D. Geometric mean

Q.10 If the median of an arrangement of numbers is equal to the mean of its middle terms then the arrangement contains

- A. Odd number of terms
- B. Even number of terms**
- C. Unlimited number of terms
- D. Prime number of terms

Q.11 In an ordered array of data having even number of values, median is the _____

- A. Sum of 2 middle values
- B. Average of 2 middle values**
- C. Average of 4 middle values
- D. Differences of 2 middle values

Q.12 Mode of a set of values can be

- A. Exactly one B. Two C. Three
D. More than one

Q.13 Bivariate categorical data can be organized as

A. The Summary Table

B. Contingency Tables

C. Side by Side Bar charts

D.

Only option 2 & 3

Tabulating and Graphing Bivariate Categorical Data

Bivariate categorical data can be organized as :

- Contingency Tables
- Side by Side Bar charts

Q.14 The formula of mean is

A. Harmonic mean

B.

Geometric mean

C. Arithmetic mean

D.

Mid range

Q.15 Average is calculated as _____

A. Sum of observation / Number of observation

B. Number of observation / Sum of observation

C. Sum of observation - Number of observation

D. Sum of observation + Number of observation

Q.16 The average of the first and third quartiles is called _____

A. Range B. Midrange C.
Interquartile range **D. Midhinge**

MIDHINGE:

Midhinge is the average of the first and third quartiles.

$$\text{Midhinge} = \frac{Q1 + Q3}{2}$$

Q.17 Class width of the given data is defined as ratio of

- A. Range and number of values in the given data
- B. Range and the number of classes**
- C. Range and midpoint/ value
- D. None of these

In our example

$$\text{Class width} = \frac{\text{Range}}{\text{Number of classes}} = \frac{46}{5} = 9.2$$

Round up 9.2 to 10

Q.18 Central tendency is tendency of data values to _____ in the central part of data.

A. **Cluster** B. Disperse C. Both
of these D. None of these

Q.19 Which of the following statements is true for Positively Skewed Distribution?

A. Mean > median > mode

B. Mean = median = mode

C. Mode > median = mean

D.

None of given

Positively Skewed Distribution
(Tilted to left)

mean > median > mode

Q.20 For the class 165 – 184 the mid point will be

A. 172.5

B. 174.5

C. 176.5

D. To find mid point at least two classes are required

Q.21 Truncated mean is also called

- A. Arithmetic mean **B. Trimmed mean** C. Harmonic mean D. None of these

Q.22 Trimmed mean is the mean of _____

A. The remaining data

B. The middle data

C. The data from last

D. None of these

Q.23 In the arrangement of numbers, by following stem and leaf format a number say '27' is located by taking

- A. 2 as its leaf and 7 as its stem
- B. 2 as its stem and 7 as its leaf**
- C. 27 as its stem with nil leaf
- D. Nil as its stem 27 as a leaf

Q.24 Fifth decile for 10.1, 10.3, 10.5, 10.6, 10.7, 10.9 is approximately _____

- A. 10.1 B. 10.3 C.
10.6 D. 10.9

DECILES

Deciles divide data into 10 equal parts

Syntax

1st Decile $D_1 = (n+1)/10$

2nd Decile $D_2 = 2(n+1)/10$

9th Decile $D_9 = 9(n+1)/10$

Q.25 In stem and leaf display stem shows

- A. Decimal number
- B. Unit number
- C. Thousandth number
- D. None of these**

This is represented in the plot as a stem of 2 and a leaf of 1. The digit at the tenth place is taken as stem and the digit at units place is taken as leaf. Similarly, 26 is represented in the plot as a stem of 2 and a leaf of 6.

Remember, a stem is displayed once and the leaf can take on the values from 0 to 9.

Q.26 Geometric mean of 130, 140, 160 is

- A. 130.5 B. 135.5 **C.**
142.8 D. 145.7

GEOMETRIC MEAN

Geometric mean is defined as the root of product of individual values. Typical syntax is as under:

$$G = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n}$$

Example

Find GM of 130, 140, 160

$$\begin{aligned} \text{GM} &= (130 \cdot 140 \cdot 160)^{1/3} \\ &= 142.8 \end{aligned}$$

Q. 27 The statement

“Median does not get affected by extreme values or outliers in the data” is _____

A. True

B. False

Q.28 Which of the following statements is true for Symmetrical Distribution?

A. Mean $>$ median $>$ mode

B. Mean = median = mode

C. Mode $>$ median $>$ mean

D.

None of given

Symmetrical Distribution
mean = median = mode

Q.29 Which of the following is true?

A. Mean is better than Median but not to Mode

B. Mean is better than Mode but not to Median

C. Mean is like as Median and Mode

D. Mean is better than both Median and Mode

Q.30 Arrangement of numbers:
3,5,8,10,12,15,16 has

- A. '10' as its mode
- B. No mode**
- C. All the terms can be taken as mode
- D. '11' as its mode

Q.31 Which of the following is true?

A. $A.M > G.M > H.M$

B. $A.M < G.M < H.M$

C. $A.M = G.M = H.M$

D. None of these

$$AM \geq GM \geq HM.$$

It is easy to remember noting that the alphabetical order of the letters A , G , and H is preserved in the inequality.

The higher exponent in the formula of average power, the greater the value of the average

Q.32 The statement that the relative frequency can be converted into cumulative frequency by adding the current frequency to the previous total

A. True

B. False

Q.33 The statement

“Mode does not get affected by extreme values or outliers in the data” is _____

A. True

B. False

Q.34 If the number of data elements is 10 and mean of the data is 50 then the sum of the data elements will be

A. 5

B. 150

C. 60

D. None of these

$$\frac{500}{10} = 50$$

Q.35 Frequency Function calculates how often values occur within a range of values.

A. True

B. False

Q.36 The median of ungroup data is:

- A. Middle value, if values are even
- B. **Middle value, if values are odd**
- C. Both above
- D. None of these

Q.37 The statement

“Mean does not get affected by extreme values or outliers in the data” is _____

A. True

B. False

Q.38 Which of the following is NOT a measure of central tendency?

- A. Percentile B. Quartile C.
Standard deviation D. Mode

Q.39 The median of the numbers 2,4,7,12 is

A. 4 **B. 5.5** C. 7 D. There
is no median for this data

Q.40 The depth of a river at four different points is 2,7,5,6 feet then how long people can cross it?

A. 7

B. 6

C. 5

D. 4.5

Q.41 Under which of the following conditions for three terms a , b , c their all central tendencies (mean, median & mode) are equal

- A. $a=b=c$** B. $a>b>c$ C.
 $a<b<c$ D. $a>b$ but $b<c$

Q.42 The mode of the data 2.7, 3.5, 4.9, 5.1, 8.3 is

- A. 3.5 B. 4.9 C. 5.1
D. There is no mode for the data

Q.43 Frequency polygon is a obtained from a frequency distribution.

- A. Bar chart **B. A line graph** C. Pareto Chart D. None of these

Q.44 The mode of the data 6,6,9,9,9,10,11,11,15,18 is

A. 6 **B. 9** C. 11 D. There
is no mode for this data

Q.45 Midhinge is same as

- A. Semi interquartile range B. Median
C. Mid Range **D. None of these**

Q.46 Linear Graphs are obtained from

- A. Linear equation** B. Quadratic
equations C. Cubic equation D. None
of these

Q.47 The Harmonic Mean of the data 8, 5, 3 is:

- A. 3.44 **B. 4.55** C. 5.44
D. None of these

Q.48 Different ways of organizing univariate categorical data are

A. The Summary Table B. Bar and Pie Charts, the Pareto Diagram

C. Contingency Tables

Tabulating and Graphing Univariate Categorical Data

There are different ways of organizing univariate categorical data:

- The Summary Table
- Bar and Pie Charts, the Pareto Diagram

Q.49 The average of smallest and largest values in the data is called _____

- A. Range **B. Midrange** C.
Interquartile range D. Midhinge

MIDRANGE

Midrange is the average of smallest and largest value. In other words it is half of a range. Midrange is affected by extreme values as it is based on smallest and largest values

Q.50 For Symmetrical Distribution, if mean = 20, then _____

- A. Mode = 19. Median = 18
- B. Mode = 18. Median = 18
- C. Mode = 20. Median = 20**
- D. Mode = 19. Median = 19

Q.51 For a frequency distribution of a data we have a significant class interval 50-54 then its mid point is

A. 2

B. 27

C. 25

D. 52

Q.52 If the value '2' collected in the data $n/2$ times and '3' occurs $n/2$ -time then the arithmetic mean is

- A. $5n/2$ B. $5/2$ C. $5(n-1)/2$
D. None of these

Q.53 Which of the following is true?

A. Arithmetic mean is based on all and Mode is on the middle values.

B. Median based on most frequent value and mode on middle value.

C. Arithmetic mean is based on all and median is on the middle values

D. None of these

Q.54 If the mean between four numbers 2, x, 3, y is '3' then the mean of 'x' and 'y' is

- A. 7 B. 1.50
C. 2 **D. 3.50**

$$\frac{12}{4} = \text{mean } 3, \text{ so } x + y = 7 \text{ and } \frac{7}{2} = 3.5$$

Q.55 The formula for finding the geometric mean of the data $(x_1, x_2, x_3, \dots, x_n)$ is given by

A. $\bar{X} = (X_1 + X_2 + X_3 + \dots + X_n) / n$

B. $\bar{X} = (X_1 \times X_2 \times X_3 \times \dots \times X_n)^{1/n}$

C. $\bar{X} = (X_1 \times X_2 \times X_3 \times \dots \times X_n)^n$

D. $\bar{X} = (X_1 + X_2 + X_3 + \dots + X_n)^{1/n}$